



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Name of Student: Aryan Singh Kasera

PRN: 25070122052

Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 13

TITLE :Write a Java program to perform reading from and writing to a file using file handling operations.

Problem Statement

Write a Java program to perform reading from and writing to a file using file handling operations.

Learning Objectives

To perform file reading and writing operations using Java file handling classes.

Theory

Java provides file handling through the java.io package. Classes such as File, FileReader, FileWriter, BufferedReader, and BufferedWriter are used to perform input and output operations. File handling enables permanent storage and retrieval of application data. Proper exception handling is necessary to manage file-related errors. These concepts are essential for data persistence.

Create a program to write and read **employee details from a file**.

```
library > Aryan > J EmployeeFile.java > Java > EmployeeFile > main(String[] args)
1  import java.io.*;
2
3  public class EmployeeFile {
    Run | Debug | Run main | Debug main
4      public static void main(String[] args) {
5
6          try {
7              // Writing to file
8              FileWriter fw = new FileWriter(fileName: "employee.txt");
9
10             fw.write(str: "Employee ID : 052\n");
11             fw.write(str: "Name      : Aryan Singh Kasera\n");
12             fw.write(str: "Department : Computer Science\n");
13             fw.write(str: "Salary    : 70000\n");
14
15             fw.close();
16
17             System.out.println(x: "Employee details written successfully.\n");
18
19             // Reading from file
20             FileReader fr = new FileReader(fileName: "employee.txt");
21             BufferedReader br = new BufferedReader(fr);
22
23             String line;
24
25             System.out.println(x: "Employee Details:");
26
27             while ((line = br.readLine()) != null) {
28                 System.out.println(line);
29             }
30
31             br.close();
32
33         } catch (IOException e) {
34             System.out.println("Error: " + e.getMessage());
35         }
36     }
37 }
```

OUTPUT

```
Employee Details:
Employee ID : 052
Name      : Aryan Singh Kasera
Department : Computer Science
Salary    : 70000
```

Create a Java program to write **patient details** (patient ID, name, age, and diagnosis) to a file and read the details from the file for display.

```
Welcome EmployeeFile.java 4 PatientFile.java 2 x
library > Aryan > J PatientFile.java > Java > PatientFile > main(String[] args)
1  import java.io.*;
2
3  public class PatientFile {
4
5      Run | Debug | Run main | Debug main
      public static void main(String[] args) {
6
7          try {
8              // Writing to file
9              FileWriter writer = new FileWriter(fileName: "patient.txt");
10
11              writer.write(str: "Patient ID : 201\n");
12              writer.write(str: "Name      : Aryan Singh\n");
13              writer.write(str: "Age       : 20\n");
14              writer.write(str: "Diagnosis  : Fever\n");
15
16              writer.close();
17
18              System.out.println(x: "Data written successfully.\n");
19
20              // Reading from file
21              BufferedReader reader = new BufferedReader(new FileReader(fileName: "patient.txt"));
22
23              String line;
24
25              System.out.println(x: "Patient Details:");
26
27              while ((line = reader.readLine()) != null) {
28                  System.out.println(line);
29              }
30
31              reader.close();
32
33          } catch (IOException e) {
34              System.out.println("Error: " + e.getMessage());
35          }
36      }
37  }
```

OUTPUT

```
Patient Details:  
Patient ID : 201  
Name      : Aryan Singh  
Age       : 20  
Diagnosis  : Fever
```

Conclusion

The program successfully demonstrates file handling in Java by writing data to a file and then reading it back. It uses classes from the `java.io` package such as `FileWriter`, `FileReader`, and `BufferedReader` along with exception handling to perform safe input and output operations. This experiment shows how Java can be used to store and retrieve data permanently using files.